

TABLE II
Exposure Limits for
RF Fields (900MHz)

Limit	Country
0,000.45 W/m ²	BUND recommendation 1997
0,001 W/m ²	Precautionary limit in Austria
0,02 W/m ²	Exposure limit in Russia
0,045 W/m ²	ECOLOG-recommendation 1998 (Germany)-Switzerland
0,1 W/m ²	Exposure limit in Poland
0,16 W/m ²	Exposure limit in Italy, Belgium
0,24 W/m ²	Exposure limit in CSSR
2 W/m ²	Exposure limit in New Zealand
3 W/m ²	Exposure limit in Canada (Safety Code 6, 1997)
4.5 W/m ²	Exposure limit in Germany, India as per ICNIRP recommendation 1998

TABLE III
Exposure Limits for
RF Fields (1800MHz)

Limit	Country
0,000.9 W/m ²	BUND recommendation 1997
0,001 W/m ²	Precautionary limit in Austria
0,02 W/m ²	Exposure limit in Russia
0,09 W/m ²	ECOLOG-recommendation 1998 (Germany)-Switzerland
0,1 W/m ²	Exposure limit in Poland
0,16 W/m ²	Exposure limit in Italy, Belgium
0,24 W/m ²	Exposure limit in CSSR
2 W/m ²	Exposure limit in New Zealand
3 W/m ²	Exposure limit in Canada (Safety Code 6, 1997)
9 W/m ²	Exposure limit in Germany and India based on ICNIRP recommendation 1998

(rather than measuring SAR). Here workers refer to maintenance and installation staff. The threshold limits adopted by ICNIRP (for workers and

TABLE IV
Exposure Limits for
RF Fields (>2000MHz)

Limit	Country
10 W/m ²	Exposure limit in India and ICNIRP recommendation 1998

Value applicable at 20,000 MHz to 300 GHz. Covers 2.1GHz band of 3G spectrum

TABLE V
E-field Reference Levels

Frequency range	E-field strength (V/m)
1-25 Hz	10000
0,025-3 kHz	250/f
0,003-1 MHz	87
1-10 MHz	87/√f
10-400 MHz	28
400-2000 MHz	1,375 √f
2-300 GHz	61

general public) for cellphone frequencies of interest are shown in the form of graph and table in Figs 14 and 15, respectively.

The reference levels recommended by ICNIRP have been adopted by India and several other countries. Guidelines based on ICNIRP-specified levels have been issued by the Ministry of Communication & IT for compliance by service providers based on the recommendations of Telecom Regulatory Authority of India (TRAI). Relevant test and measurement procedures of EMR have been issued by the Telecommunication Engineering Centre (TEC).

The power density limits adopted by some of the other nations are cited in Tables II through IV (some figures may be approximate). From these exposure limits, it is observed that standards adopted by east European countries ($P_D \leq 1 \text{ W/m}^2$) and few others are based on precautionary approaches. The crusaders against the radiation hazards from mobile towers accordingly want the Indian government to adopt precautionary standards rather than ICNIRP-specified

limits, which are amongst the worst in the world.

EMR measurements

Generally, the unit used for characterising the total electromagnetic field is the 'power density'. Power density is most appropriately used when the point of measurement is far away from an antenna to be located in the far-field zone of the antenna (say, 10 metres or more from the antenna). Since power density (P_D) can be easily derived from electric field intensity ('E' in RMS volt per metre) using equation $P_D = E^2/377 \text{ W/m}^2$, we may instead measure the electric field intensity and derive power density value therefrom.

An important point worth noting is that if multiple field sources (TV, FM or other cell towers) exist, special procedures must be used. Multiple sources may include near- and far-field conditions. Performing correct measurements requires consideration of frequency, polarisation, modulation, and 'on' and 'off' times of each source. Interference from other sources outside the frequency range that the instrument is designed to measure can greatly degrade measurement accuracy. This interference, called out-of-band interference, is important in areas where multiple signals are present. Simultaneous exposures to fields of different frequencies are additive in their effects. Hence to determine that compliance limits are not exceeded, ensure that E-field strength is measured at all relevant frequencies and summation done using the following equation:

$$\sum_{i=1}^{n} \left(\frac{E_i}{E_{ref,i}} \right)^2 \leq 1$$

$i = 100 \text{ kHz} \quad i > 1 \text{ MHz}$

where 'E_i' is the electric field strength measured at frequency 'i', 'E_{ref,i}' is the corresponding E-field reference value from Table V (based on ICNIRP-recommended E-field levels) and C = 87/√f V/m.

If the overall sum is less than unity, compliance limits are not exceeded.